

RESEARCH ARTICLE

Precise Electronic Modulation of Thianthrene-Cored Covalent Organic Frameworks for High-Efficiency H₂O₂ Electrosynthesis in Neutral Media

Xiaoyu Xu^{1,2,3} | Yihe Chen⁴ | Shuang Zheng^{2,3} | Qingqing Cheng⁴  | Yue Wang^{2,3} | Shijie Ma^{2,3} | Yanhong Xu¹ | Qing Xu^{2,3} | Gaofeng Zeng^{2,3} 

¹Key Laboratory of Preparation and Applications of Environmental Friendly Materials, Key Laboratory of Functional Materials Physics and Chemistry of the Ministry of Education, Jilin Normal University, Changchun, Jilin, China | ²Advanced Separation & Conversion on Engineered Nanopore Dynamics Laboratory, Center For Low-carbon Conversion Science and Engineering, Shanghai Advanced Research Institute, Chinese Academy of Sciences, Shanghai, China | ³School of Chemical Engineering, University of Chinese Academy of Sciences, Beijing, China | ⁴Shanghai Advanced Research Institute, Chinese Academy of Sciences, Shanghai, China

Correspondence: Yanhong Xu (xuyh@jlnu.edu.cn) | Qing Xu (xuqing@sari.ac.cn) | Gaofeng Zeng (zenggf@sari.ac.cn)

Received: 9 January 2026 | **Revised:** 30 March 2026 | **Accepted:** 8 April 2026

Keywords: 2e⁻ oxygen reduction reaction | covalent organic frameworks | electrocatalysis H₂O₂ | metal-free catalyst | neutral conditions

ABSTRACT

Covalent organic frameworks (COFs) have emerged as tunable electrocatalysts for hydrogen peroxide synthesis via two-electron oxygen reduction in alkaline media. However, inherent H₂O₂ decomposition in such environments necessitates the development of neutral-pH alternatives, which remains an unresolved challenge for COF-based systems. Herein, this work pioneers thianthrene-integrated COF catalysts for efficient H₂O₂ electrosynthesis under neutral conditions. By engineering linker architectures, we constructed three COFs with comparable crystallinity, porosities, and stability, but distinct electronic states. The optimized TFPTA-TTPE-COF exhibits high activity with a mass activity of 11.8 A g⁻¹ and the H₂O₂ selectivity of 93.8%. Importantly, flow-cell system validation demonstrates industrial-scale productivity, achieving 4461 mmol g⁻¹ h⁻¹ H₂O₂ yield at a faradaic efficiency of 79.7% under 150 mA cm⁻². Theoretical analyses establish carbon atoms within imine linkers as primary active sites, where subtle structural variations fine-tune OOH* intermediate binding energies to govern activity and selectivity. This work provides fundamental insights into designing active COFs electrocatalysts for sustainable H₂O₂ production.

1 | Introduction

Hydrogen peroxide (H₂O₂) serves as a pivotal reagent used in chemical synthesis, bleaching, wastewater treatment, disinfection, and energy technologies [1–5]. Traditional industrial production methods, such as the anthraquinone process, remain prevalent but suffer from high costs, operational complexity, and environmental pollution [6–8]. In contrast, electrocatalytic synthesis of H₂O₂ presents a green and highly

efficient alternative. To date, a variety of catalysts have been developed, including precious metal alloys [9], non-precious metal clusters or atoms [10–13], and metal-free carbons [14]. Among them, metal-free catalysts attract particular interest due to their low cost and poison resistance [15–17]. However, conventional carbon-based metal-free catalysts generally possess ill-defined active sites and heterogeneous structures, which hinder precise modulation of their catalytic performance [18, 19].

Xiaoyu Xu and Yihe Chen contributed equally to this work.

Covalent organic frameworks (COFs), crystalline porous polymers constructed from organic building blocks via covalent bonds [20–22], have emerged as a promising class of functional materials. Various functional COFs have demonstrated utility in gas storage and separation, optoelectronics, catalysis, and energy storage [23–29]. Their well-defined pore architectures facilitate mass transport, and their functionalized backbones allow for precise catalytic site design, positioning COFs as promising candidates for electrocatalysis, including carbon dioxide reduction reaction (CO₂RR) [30–33], oxygen evolution reaction (OER) [34, 35], hydrogen evolution reaction (HER) [36, 37], and oxygen reduction reaction (ORR) [38–41]. Notably, metal-free COFs have recently emerged for H₂O₂ electrosynthesis via two-electron (2e[−]) ORR [42, 43]. Wang et al. designed a thiophene-rich fully conjugated 3D COF (BUCT-COF-7) via a facile group-regulation strategy, which enables the rational tuning of the ORR catalytic pathway and thus achieves outstanding 2e[−] ORR activity [44]. Long et al. implemented oligoether chain (EO) engineering by embedding oligoether segments with different lengths into COFs, constructing a metal-free COF-based electrocatalyst with high reactivity for the ORR toward H₂O₂ electrosynthesis [45]. Furthermore, a variety of modification approaches have been introduced, such as fluorinated modifications in PYTA-TPEDH-COF, edge-site engineering in 1D TAPB-TA-COF, and benzyl-functionalized in JUC-660 to switch the ORR pathway from 4e[−] to 2e[−] [46–48]. However, all these systems operate exclusively in alkaline media (0.1 M KOH), where H₂O₂ readily decomposes [49–52]. Thus, the development of metal-free COF catalysts capable of efficient 2e[−] ORR in neutral electrolytes remains an urgent and challenging goal.

Herein, we have demonstrated the construction of catalytic COFs for neutral synthesis of H₂O₂ by employing thianthrene-based cores. By linking thianthrene cores with different conjugated building blocks, three COFs, TFPTA-PY-COF, TFPTA-TPE-COF, and TFPTA-TTPE-COF, were synthesized with identical *sql* topology and comparable porosity. Systematic variation of conjugated linkers modulated charge transfer kinetics and electron-donating capacities, directly influencing catalytic performance. TFPTA-TTPE-COF demonstrated higher selectivity for 2e[−] ORR in neutral conditions compared to TFPTA-PY-COF and TFPTA-TPE-COF, due to the easier formation of intermediates OOH*. Importantly, flow-cell testing demonstrated the synthesis of H₂O₂ at a rate of 4461.4 mmol g^{−1} h^{−1} at a high current density of 150 mA cm^{−2}, offering a promising candidate and valuable reference for sustainable hydrogen peroxide electrosynthesis.

2 | Results and Discussion

2.1 | Design and Properties of Thianthrene-Based COFs

Three thianthrene-core COFs, TFPTA-PY-COF, TFPTA-TPE-COF, and TFPTA-TTPE-COF, were constructed by condensing 4,4',4'',4'''-(thianthrene-2,3,7,8-tetrayl) tetrabenzaldehyde (TFPTA) with amine linkers of 4,4',4'',4'''-(pyrene-1,3,6,8-tetrayl) tetraaniline (PY), tetrakis (4-aminophenyl) ethene (TPE), and tetrakis (4-aminobiphenyl) ethylene (TTPE), respectively (Figure 1). Fourier transform infrared (FT-IR) spectroscopy

confirmed the formation of the frameworks, with new bands emerging at 1626 cm^{−1} for TFPTA-PY-COF and 1632 cm^{−1} for TFPTA-TPE-COF and TFPTA-TTPE-COF, attributable to the characteristic C=N linkages. Concurrently, the C–S bond peak at 1172 cm^{−1} across all spectra confirmed retention of the thianthrene structural integrity (Figures S1–S3).

Powder X-ray diffraction (PXRD) measurements were conducted to explore the crystal structures of COFs. TFPTA-PY-COF revealed characteristic reflection peaks at 5.13, 10.19, 13.21, 14.43, 20.69 and 25.81°, corresponding to the (110), (2-20), (3-10), (320), (4-40), and (1-3-1) facets, respectively (Figure 2a), and Pawley refinement demonstrated excellent agreement between AA stacking theoretical structures and experimental results (R_{wp} of 6.34% and R_p of 4.62%) (Table S1). AA stacking was confirmed with cell parameters of a = 20.31 Å, b = 30.29 Å, c = 3.74 Å, α = 89.9°, β = 88.59°, γ = 91.9°. Similarly, TFPTA-TPE-COF displayed prominent peaks at 3.40° (100), 7.51° (210), and 20.80° (001), while TFPTA-TTPE-COF exhibited distinct (100) and (110) reflections (Figure 2b,c). Pawley refinements indicated that they adopted eclipsed AA models rather than AB staggered stacking models with R_{wp}/R_p = 4.15%/3.17% for TFPTA-TPE-COF and 3.46%/2.48% for TFPTA-TTPE-COF (Tables S2 and S3).

Nitrogen (N₂) physisorption at 77 K exhibited similar type-I isotherms for all COFs, suggesting their similar microporous channels (Figure 2d–f). Brunauer-Emmett-Teller (BET) surface areas for TFPTA-PY-COF, TFPTA-TPE-COF, and TFPTA-TTPE-COF were determined to be 807.6, 791.6, and 580.5 m² g^{−1}, with corresponding pore volumes of 0.90, 0.96, and 0.55 cm³ g^{−1}, respectively. Pore size distribution curves revealed that the pore sizes in three COFs were close to the theoretical predictions (Figure 2g–i; Figures S4–S6).

Scanning electron microscopy (SEM) and transmission electron microscopy (TEM) revealed consistent granular morphologies across all COFs, exhibiting uniform rice-like stereoscopic architectures (Figures S7–S9). Energy dispersive X-ray spectroscopy (EDS) mapping confirmed uniform distribution of elements (C, N, and S) throughout the COF skeletons, thus verifying structural integrity (Figures S10–S12). High-resolution TEM imaging further corroborated crystallinity through well-defined lattice fringes. TFPTA-PY-COF exhibited distinct lattice fringes with an interspacing of 1.74 nm, corresponding to the (110) facet. Similarly, TFPTA-TPE-COF and TFPTA-TTPE-COF displayed lattice spacings attributable to their respective (100) facets (Figure 2j–l; Figures S13–S15).

Solid-state ¹³C nuclear magnetic resonance (NMR) spectra and X-ray photoelectron spectroscopy (XPS) elucidated the chemical architectures of the synthesized COFs. ¹³C NMR spectra confirmed imine bond (C=N) formation at 160 ppm for TFPTA-PY-COF with C–S at 138 ppm (Figure 3a). Corresponding carbon environments were unequivocally identified in TFPTA-TPE-COF and TFPTA-TTPE-COF, validating structural consistency. XPS analysis detected the presence of C, N, and S in COFs, aligning with EDS mapping. High-resolution S 2p spectra displayed peaks at 164.8 and 163.5 eV, attributed to S 2p_{3/2} and S 2p_{1/2}, confirming intact thianthrene units (Figure 3b). N 1s spectra of all COFs further evidenced C–N and C=N,

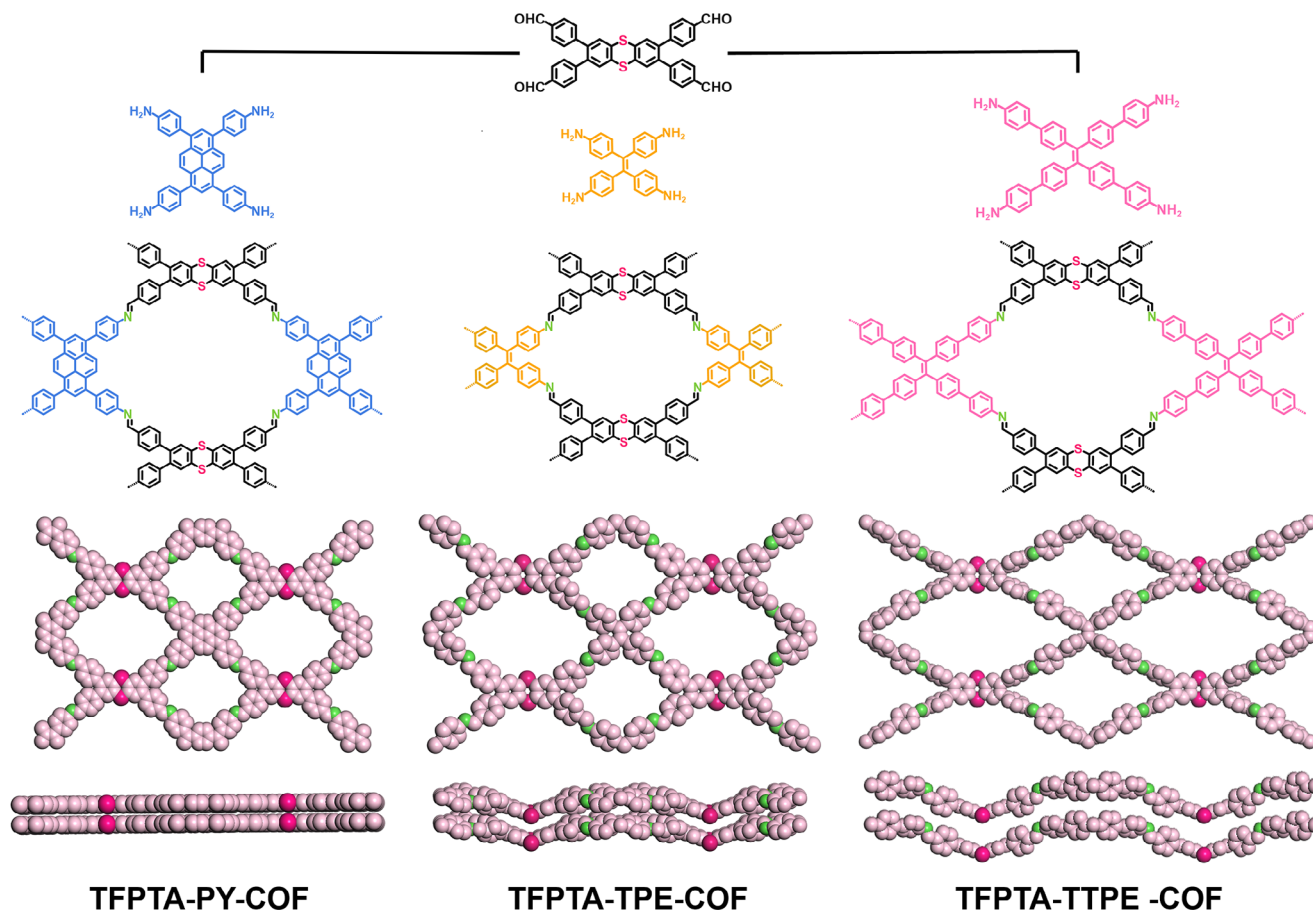


FIGURE 1 | Synthetic route (upper) and top/side views (down) of the AA stacking structure of TFPTA-PY-COF, TFPTA-TPE-COF, and TFPTA-TTPE-COF.

indicating the successful formation of the COFs (Figures S16 and S17).

To further explore the influence of linkers on the electronic states of the COFs, their energy levels were calculated via ultraviolet-visible spectroscopy (UV-vis) and Mott-Schottky measurements. All COFs showed strong absorption peaks between 200 and 600 nm (Figure 3c). Tauc plots indicated band gaps of 2.34, 2.27, and 2.37 eV for TFPTA-PY-COF, TFPTA-TPE-COF, and TFPTA-TTPE-COF, respectively (Figure S18). Mott-Schottky measurements revealed the highest occupied molecular orbital (HOMO) and lowest unoccupied molecular orbital (LUMO). The LUMO levels of TFPTA-PY-COF, TFPTA-TPE-COF, and TFPTA-TTPE-COF were -2.97 , -0.75 , and -0.16 eV, while the HOMO levels were -5.31 , -3.02 , and -2.53 eV, respectively, indicating that TFPTA-TTPE-COF had a stronger reductive ability than TFPTA-TPE-COF and TFPTA-PY-COF (Figure 3d; Figure S19).

The thermal and chemical stabilities of the synthesized COFs were evaluated. Thermogravimetric analysis under N_2 atmosphere revealed excellent thermal stability, with significant mass loss commencing only above $500^\circ C$ for all COFs. Even under an air atmosphere, structural decomposition initiated beyond $450^\circ C$ (Figure S20). Chemical stability assessments involved immersion in water and $0.1 M K_2SO_4$ solutions for 7 days. Post-treatment characterization through PXRD, FT-IR, XPS, and N_2 physisorption demonstrated preserved crystallinity, functional

group retention, and elemental composition consistency (Figures S21–S26). Minor PXRD shift ($\sim 0.2^\circ$) in primary reflections was attributed to guest molecule (H_2O/THF) residence within pores. This induced framework expansion through bulking of flexible imine linkages, consistent with dynamic structural responses observed in COFs [53–56].

2.2 | Design and Properties of Thianthrene-Based COFs

The electrochemical performance of the COFs was assessed using the standard rotating ring-disc electrode (RRDE) technique in a $0.1 M K_2SO_4$ solution with a three-electrode system, where the COFs served as the working electrodes. Initial cyclic voltammetry (CV) measurements were conducted at a scan rate of $50 mV s^{-1}$ in N_2 or O_2 -saturated $0.1 M K_2SO_4$ aqueous solutions. Notably, obvious reductive peaks under O_2 atmosphere were observed for all COFs, indicative of their catalytic activity toward ORR (Figure S27).

The catalytic activities and selectivity of the COFs were reflected by the current densities obtained from the disk and ring electrodes. The onset potentials (at current density of $0.1 mA cm^{-2}$) for TFPTA-PY-COF, TFPTA-TPE-COF, and TFPTA-TTPE-COF were around 0.40 , 0.41 , and $0.39 V$ vs. RHE, respectively (Figure 4a). TFPTA-TTPE-COF exhibited the highest H_2O_2 selectivity, rang-

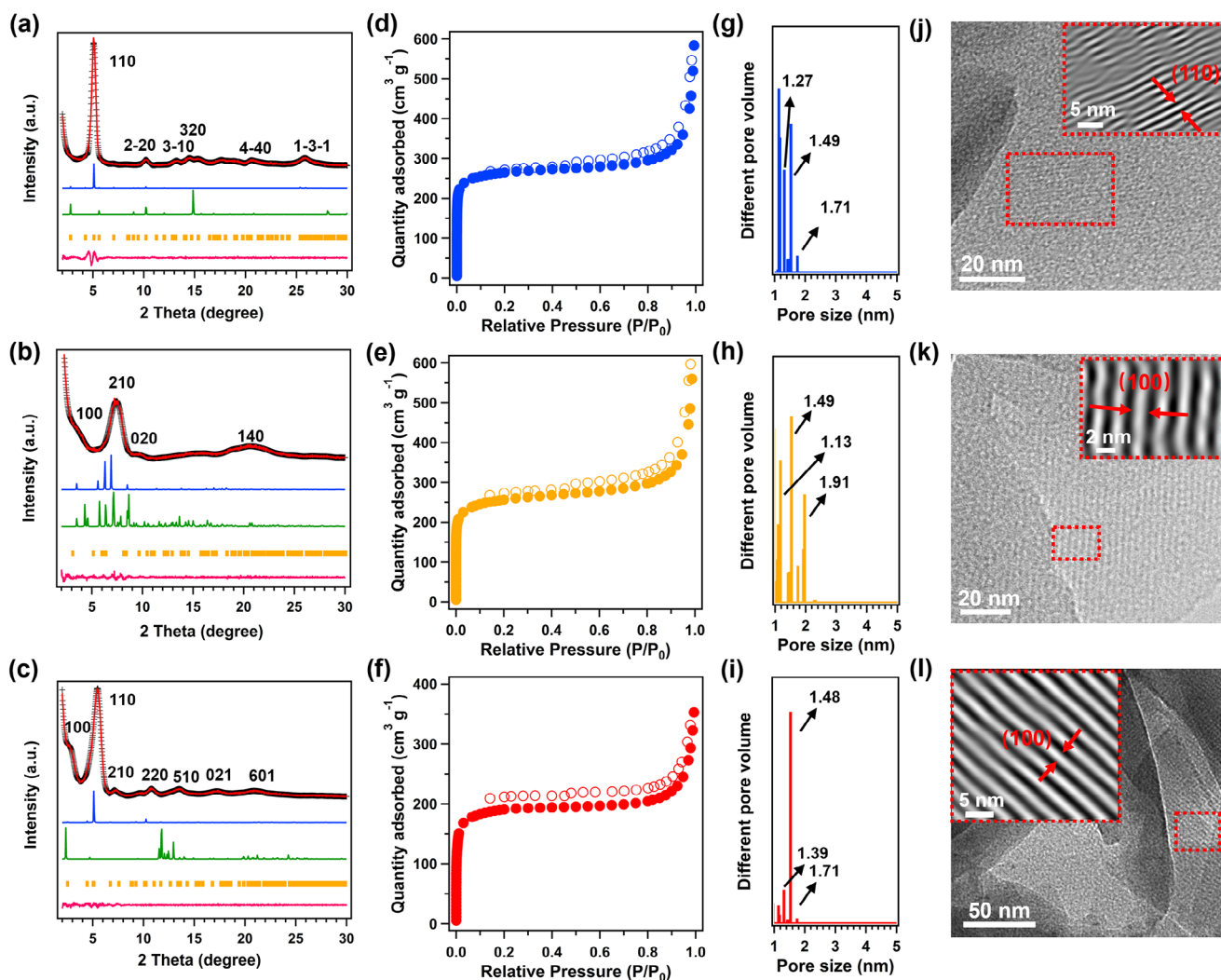


FIGURE 2 | PXRD patterns of (a) TFPTA-PY-COF, (b) TFPTA-TPE-COF, and (c) TFPTA-TTPE-COF, the experimental (red), Pawley refined (black), AA (blue), staggered AB (green) stacking modes, Bragg position (orange), and difference (pink). N₂ physisorption isotherms (d–f) and pore size distributions (g–i), and high-resolution TEM images (j–l) of (d,g,j) TFPTA-PY-COF, (e,h,k) TFPTA-TPE-COF, and (f,i,l) TFPTA-TTPE-COF.

ing from 90.5% to 93.8% at potential ranges of 0 to 0.3 V (Figure 4b, red curve). In comparison, TFPTA-TPE-COF and TFPTA-PY-COF displayed lower selectivity with corresponding H₂O₂ yields of 83.9%–89.3% (Figure 4b, orange curve), 67.2%–84.5% (Figure 4b, blue curve), respectively. The electron-transfer numbers (*n* values) were in the range of 2.19 to 2.13 for TFPTA-TTPE-COF, 2.32 to 2.23 for TFPTA-TPE-COF, and 2.65 to 2.31 for TFPTA-PY-COF (Figure 4c). Notably, the optimal H₂O₂ selectivity of TFPTA-TTPE-COF reached up to 93.8%, with an electron-transfer number of 2.13. All results were presented as the average of three independent parallel experiments, and error bars represented the standard deviation of the mean (Figure S28). Complementary alkaline-condition testing (0.1 M KOH) confirmed TFPTA-TTPE-COF still maintained the highest selectivity among the COFs, although significantly reduced relative to neutral performance (Figure S29). We further monitored H₂O₂ concentration over time in 0.1 M KOH (alkaline) and 0.1 M K₂SO₄ (neutral) electrolytes with and without COF catalysts, as shown in Figure S30. In a neutral electrolyte, the H₂O₂ concentration remained almost unchanged (~170 mM) for 108 h, similar to the pure H₂O₂ control. Differently, H₂O₂ decomposed rapidly in

alkaline solution, decreasing from ~190 to ~40 mM within 108 h. The results confirmed that neutral media were more suitable for H₂O₂ production and preservation.

Kinetic analysis via Tafel slopes revealed enhanced kinetics for TFPTA-TTPE-COF (97 mV dec^{−1}), outperforming TFPTA-TPE-COF (103 mV dec^{−1}) and TFPTA-PY-COF (108 mV dec^{−1}) (Figure 4d). In terms of catalytic activity, TFPTA-TTPE-COF achieved a turnover frequency (TOF) value of 0.072 s^{−1} and a mass activity of 11.8 A g^{−1} at a potential of 0.01 V vs. RHE, surpassing the values obtained from the other two COFs (TOF values of 0.059 and 0.056 s^{−1}, mass activities of 12.9 and 10.3 A g^{−1} for TFPTA-TPE-COF and TFPTA-PY-COF, respectively) (Figure 4e).

To reveal the contribution of the electrochemically active surface areas (ECSA) to the catalytic performance, ECSA values were determined from the electrochemical double-layer capacitances (*C_{dl}*) (Figures S31 and S32). All synthesized COFs showed similar ECSA values, with 0.44, 0.43, and 0.45 mF cm^{−2} for TFPTA-PY-COF, TFPTA-TPE-COF, and TFPTA-TTPE-COF, respectively.

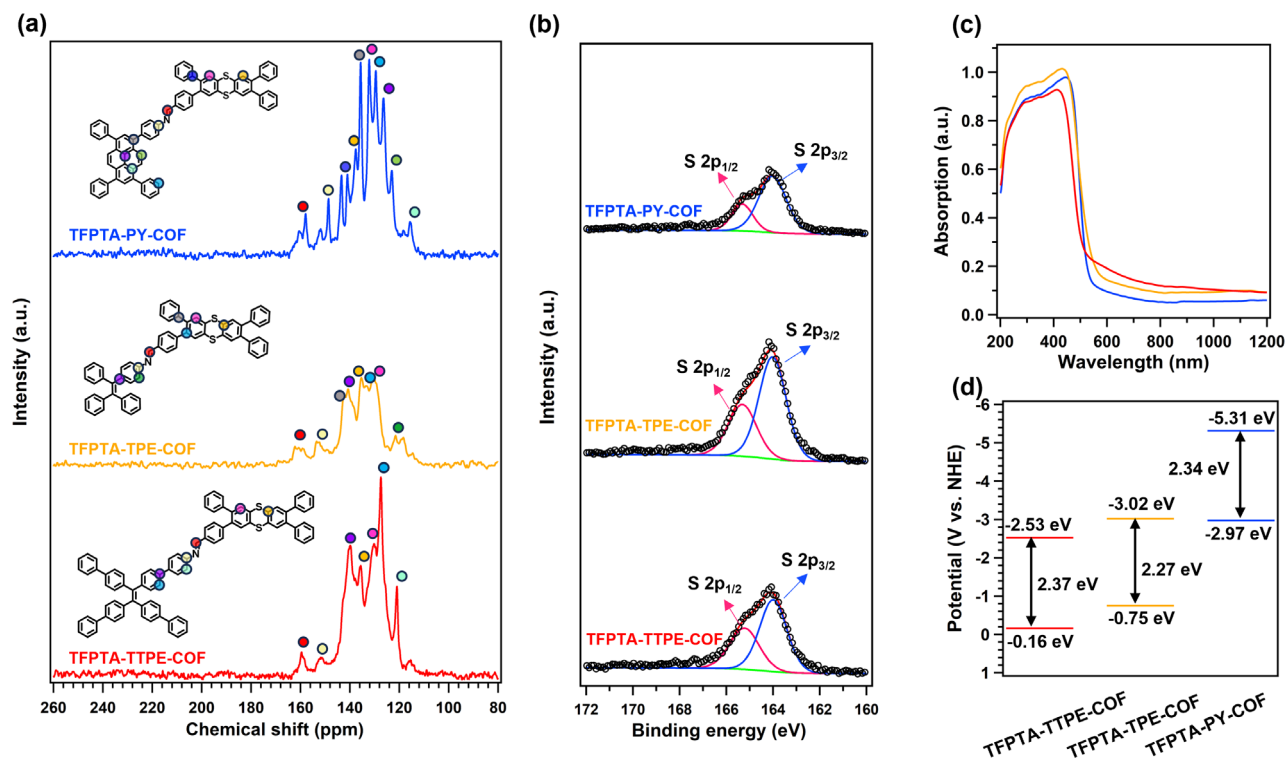


FIGURE 3 | (a) ^{13}C NMR, (b) XPS S 2p spectra, (c) solid-state UV-vis absorption spectra, and (d) band gap of TFPTA-PY-COF (blue), TFPTA-TPE-COF (orange), and TFPTA-TTPE-COF (red).

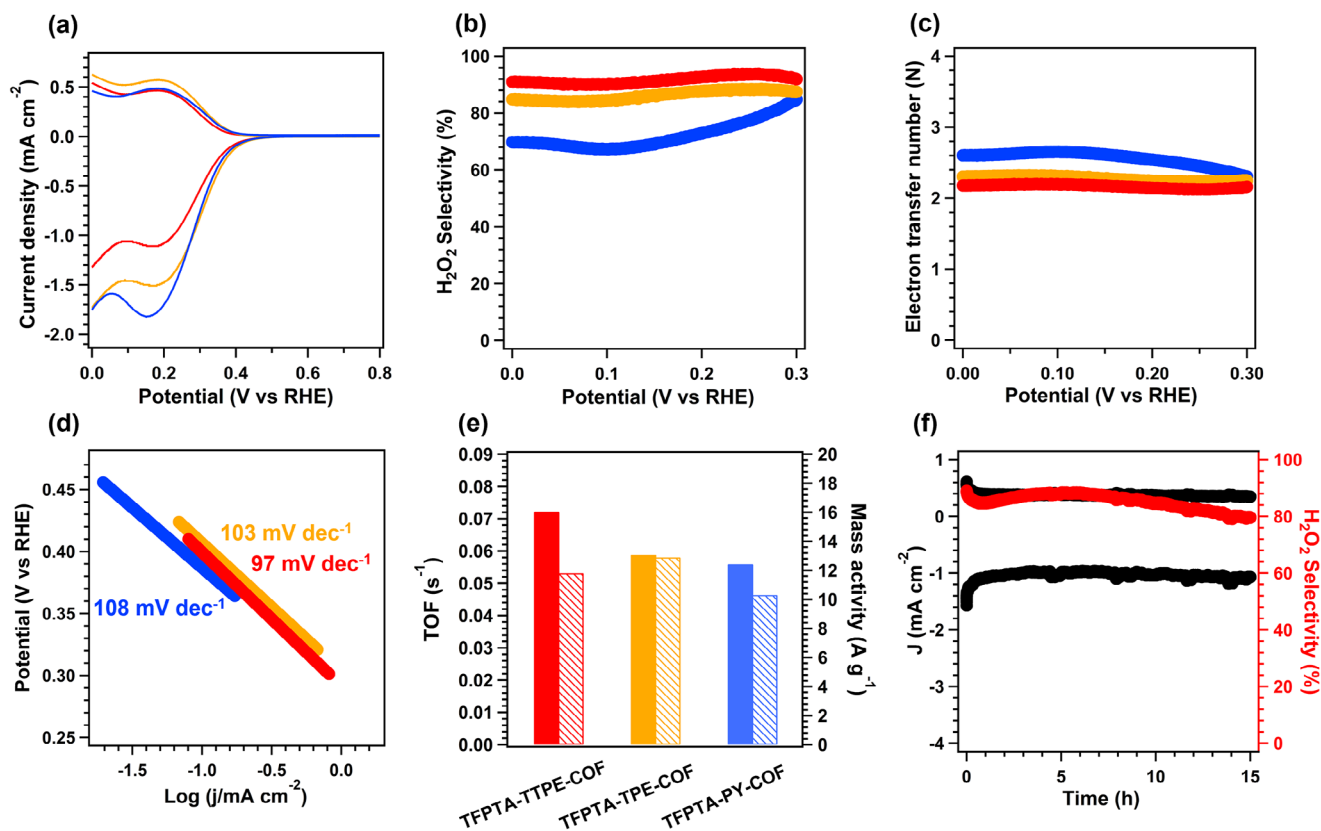


FIGURE 4 | (a) Linear sweep voltammetry (LSV) curves in O_2 -saturated 0.1 M K_2SO_4 electrolyte at 1600 rpm, (b) corresponding H_2O_2 selectivity, (c) electron transfer number, (d) Tafel slopes, and (e) Turnover frequency (TOF, solid) and mass activity values (pattern) of TFPTA-PY-COF (blue), TFPTA-TPE-COF (orange) and TFPTA-TTPE-COF (red). (f) Stability of TFPTA-TTPE-COF at a potential of 0.01 V vs RHE.

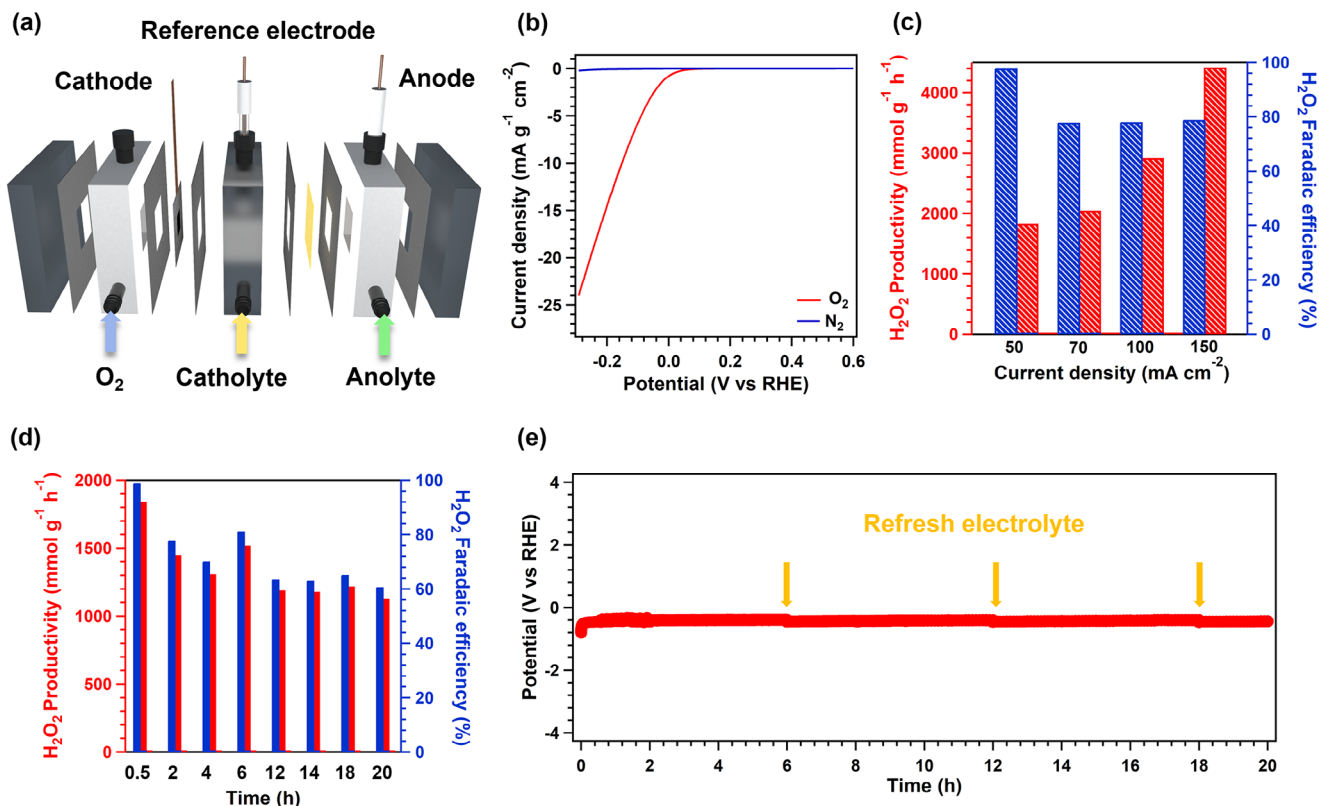


FIGURE 5 | (a) Schematic of the flow cell configuration. (b) LSV curves of TFPTA-TTPE-COF with O₂ and N₂ environments in 0.5 M K₂SO₄. Current density dependence (c) and time dependence (d) on FE (blue) and H₂O₂ yield (red) of TFPTA-TTPE-COF. (e) Stability of TFPTA-TTPE-COF in flow cell at 50 mA cm⁻² (refreshing electrolyte with 6 h interval).

Thus, the close ECSA values among the three COFs suggested that the enhanced activity of TFPTA-TTPE-COF stemmed from its intrinsic sites rather than increased exposed catalytic surface areas. Electrochemical impedance spectroscopy (EIS) further revealed reduced charge transfer resistance for TFPTA-TTPE-COF, facilitating more efficient electron transfer (Figure S33). Beyond superior activity and selectivity of TFPTA-TTPE-COF, its long-term stability testing at a potential of 0.01 V vs. RHE demonstrated reliable durability, maintaining >80% H₂O₂ selectivity over 15 h (Figure 4f).

Post-electrocatalytic structural integrity was assessed through multi-technique characterisation (Figure S34). FTIR and XPS results confirmed preservation of characteristic functional groups and unaltered elemental states and bonding environments. TEM imaging demonstrated maintained morphology without discernible structural degradation. Collectively, these investigations verify the structural robustness of COF catalysts, with no detectable compositional alterations following extended electrochemical testing.

Following the confirmation of TFPTA-TTPE-COF's outstanding selectivity for H₂O₂ production, we employed a flow cell system to evaluate its practical catalytic performance under high current densities (Figure 5a). LSV curves showed a substantial current density of 24 mA cm⁻² at -0.29 V under O₂ environment, contrasting with a mere 0.23 mA cm⁻² under N₂, thereby confirming catalytic activity originated from O₂ reduction

(Figure 5b). Subsequently, time-dependent chronoamperometry tests quantified H₂O₂ electroproduction capacity across varying current densities, with Faradaic efficiency (FE) and production rates determined via the cerium sulfate titration method (Figure S35). At current densities of 50, 70, 100, and 150 mA cm⁻², TFPTA-TTPE-COF achieved FE values of 98.9%, 78.7%, 78.9%, and 79.7%, respectively, while generating H₂O₂ at rates of 1845, 2056, 2946, and 4461 mmol g⁻¹ h⁻¹, respectively (Figure 5c). These results indicated that increasing current density elevated H₂O₂ yield while maintaining stable faradaic efficiency, highlighting robust catalytic performance under industrially relevant conditions.

To assess operational stability, continuous electrolysis on TFPTA-TTPE-COF was performed at 50 mA cm⁻² for 20 h (Figure 5d,e). Initial FE approached 100% within the first 0.5 h, yielding H₂O₂ at a rate of 1844 mmol g⁻¹ h⁻¹. After 12 h, efficiency progressively declined to a plateau of ~65%, sustaining H₂O₂ production at ~1144 mmol g⁻¹ h⁻¹ until completion of the 20 h test. Throughout this period, TFPTA-TTPE-COF maintained a sustained average production rate of 1362 mmol g⁻¹ h⁻¹, unambiguously establishing its efficacy and durability for prolonged H₂O₂ electrosynthesis. Complementary benchmarking analysis contextualised these results against state-of-the-art catalysts. Systematic comparison of electrocatalytic metrics (RRDE and flow cell performance) across recently reported COFs and metal-free catalysts confirmed the superiority of TFPTA-TTPE-COF under neutral conditions (Table S4).

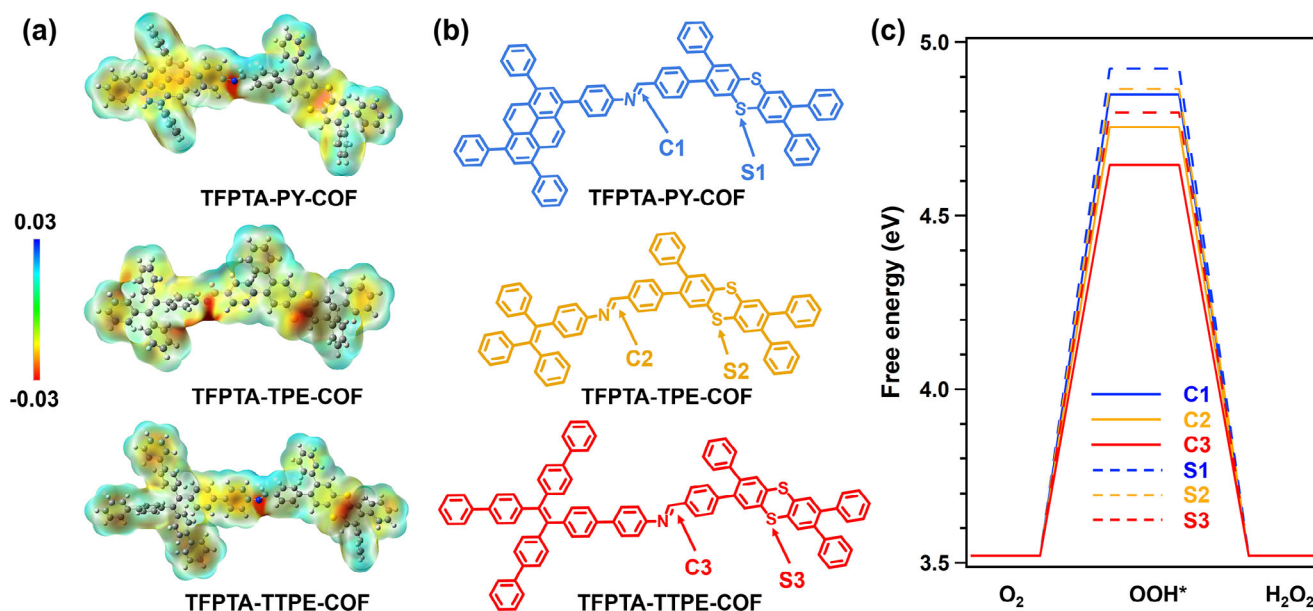


FIGURE 6 | (a) Electrostatic surface potential (ESP) maps, (b) Model structures, and (c) relative energy diagram of the 2e⁻ pathway of ORR on TFPTA-PY-COF (blue), TFPTA-TPE-COF (orange), and TFPTA-TTPE-COF (red) at a potential of 0.7 V.

2.3 | Potential Mechanism on Thianthrene-Cored COFs

To elucidate the catalytic mechanism of thianthrene-core COFs, detailed theoretical calculations were conducted. The torsion angles between PY, TPE, and TTPE units relative to the TFPTA plane were measured at 0°, 49°, and 62°, respectively, across the three COFs (Figure S36). These angular variations directly modulate π -orbital overlap along the COFs backbones, with reduced torsion enhancing π -conjugation efficiency and electron delocalisation across the 2D frameworks. Dipole moment calculations further revealed significant polarity differences, quantifying values of 2.73 eÅ⁻¹ for TFPTA-PY-COF, 2.61 eÅ⁻¹ for TFPTA-TPE-COF, and 1.54 eÅ⁻¹ for TFPTA-TTPE-COF (Figure S37). This polarity gradient would influence intermediate binding strengths during catalysis.

Bader charge analysis identified carbon atoms in C=N linkages and sulphur atoms in thianthrene units as the most electronegative sites across all COFs (Figures S38–S40), designating them as possible active sites for ORR. Electrostatic surface potential (ESP) maps and Fukui function calculations further confirmed strong electron affinity at these C=N bonded carbons and thiophene-sulfur sites (Figure 6a; Figure S41).

Density functional theory (DFT) simulations at 0.7 V vs RHE quantified free energy profiles for the 2e⁻ ORR pathway (Figure 6b). The catalytic activity depends on the formation ability of the intermediates OOH*. The free energy changes (ΔG) served as the key activity descriptor, with carbon sites exhibiting lower barriers than sulfur sites (i.e., C1 = 1.33 eV, C2 = 1.24 eV, C3 = 1.13 eV versus S1 = 1.40 eV, S2 = 1.35 eV, S3 = 1.28 eV) (Figure 6c). This confirmed that the carbon center in the C=N bond serves as the primary catalytic site, although the sulfur atom undergoes physical adsorption with O₂ via asymmetric sp² hybridization (Figure S42). Critically, the smallest ΔG value at

the C3 site correlated with the superior experimental catalytic performance of TFPTA-TTPE-COF (Figures S43–S45). Collectively, these computational results demonstrated that strategic building block selection effectively tuned electrocatalytic oxygen reduction activity for H₂O₂ synthesis under neutral conditions.

3 | Conclusion

In summary, three new thianthrene-based COFs were first synthesized for catalyzing 2e⁻ ORR in natural condition. All the synthesized COFs showed highly ordered structures and high surface areas. By altering the linkers in the COFs, their electronic properties were modulated, which influenced their catalytic activity and selectivity. The flow cell based on the optimized catalyst achieved H₂O₂ synthesis rate of 4461.4 mmol g⁻¹ h⁻¹, with FE of 79.7%. The theoretical calculation revealed that the catalytic sites were from the carbon atoms in the imine-linkages, and the higher activity for TFPTA-TTPE-COF was due to the easier formation of the intermediates. This work showed us a new exploration for COFs in energy fields.

4 | Experimental Section

4.1 | Synthesis of TFPTA-PY-COF

A Pyrex tube was charged with TFPTA (0.02 mmol, 12.7 mg), PY (0.02 mmol, 11.33 mg), 1,2-dichlorobenzene (o-DCB, 1 mL), and n-butanol (n-BuOH, 1 mL). The mixture was sonicated for 5 min to obtain a uniform solution, followed by the addition of aqueous acetic acid (0.2 mL, 6 M). The tube was frozen in a liquid nitrogen bath and degassed by three freeze–pump–thaw cycles, sealed under vacuum, and heated at 120°C for 72 h. The resulting solid was collected by filtration, washed with tetrahydrofuran, and dried under vacuum for 12 h with a yield of 86.7%.

4.2 | Synthesis of TFPTA-TPE-COF and TFPTA-TTPE-COF

TFPTA-TPE-COF was synthesized by mixing TFPTA (0.02 mmol, 12.7 mg), TPE (0.02 mmol, 7.85 mg), dimethylacetamide (DMAC, 1 mL), and 1,3,5-trimethylbenzene (Mesitylene, 1 mL) in a Pyrex tube. Following the above procedure of sonication, sealing, reaction, and washing under the same conditions, the product was obtained in 85.6% yield. The synthesis of TFPTA-TTPE-COF was carried out analogously to that of TFPTA-TPE-COF, using TTPE (0.02 mmol, 13.9 mg) in place of TPE. The product was isolated in 75.4% yield.

Acknowledgements

This work was financially supported by the National Key Research and Development Program of China (2024YFE0206900), the National Natural Science Foundation of China (52303288 and 22378413), the Youth Innovation Promotion Association of the Chinese Academy of Sciences (E324441401), the State Key Laboratory of Materials-Oriented Chemical Engineering (SKL-MCE-25B05), and the Shanghai Soft X-ray Free Electron Laser Facility beamline project.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

References

1. Z.-K. Chen, Z.-L. Shen, X.-Y. Zhang, et al., "Boosting Photocatalytic H₂O₂ Production by Fluorination of Covalent Organic Frameworks via Synergic Charge Transfer and Phase Regulation," *Applied Catalysis B: Environment and Energy* 379 (2025): 125731, <https://doi.org/10.1016/j.apcatb.2025.125731>.
2. X. Yang, Y. Zeng, W. Alnough, Y. Hou, D. Higgins, and G. Wu, "Tuning Two-Electron Oxygen-Reduction Pathways for H₂O₂ Electrosynthesis via Engineering Atomically Dispersed Single Metal Site Catalysts," *Advanced Materials* 34 (2022): 2107954, <https://doi.org/10.1002/adma.202107954>.
3. X. Wen, X. Zhang, M. Wang, et al., "Efficient Electrocatalytic H₂O₂ Activation over Nitrogen-Doped Carbon Encapsulated Co₃O₄ for Drinking Water Disinfection," *Applied Catalysis B: Environmental* 342 (2024): 123437, <https://doi.org/10.1016/j.apcatb.2023.123437>.
4. H. Chen, C. Wang, H. Wu, et al., "Host-Guest-Induced Electronic state Triggers Two-Electron Oxygen Reduction Electrocatalysis," *Nature Communications* 15 (2024): 9222, <https://doi.org/10.1038/s41467-024-53714-3>.
5. Y. Tian, M. Li, Z. Wu, et al., "Edge-Hosted Atomic Co–N 4 Sites on Hierarchical Porous Carbon for Highly Selective Two-Electron Oxygen Reduction Reaction," *Angewandte Chemie International Edition* 61 (2022): 202213296, <https://doi.org/10.1002/anie.202213296>.
6. Y. Ju, H. Lin, G. Tan, et al., "Regulating the Electronic Structure of Covalent Organic Frameworks via Heterocyclic Isomers for Highly Efficient Photocatalytic H₂O₂ Generation," *Nature Communications* 16 (2025): 5658, <https://doi.org/10.1038/s41467-025-60960-6>.
7. Q. Zhi, R. Jiang, X. Yang, et al., "Dithiine-linked Metalphthalocyanine Framework with Undulated Layers for Highly Efficient and Stable H₂O₂ Electroproduction," *Nature Communications* 15 (2024): 678, <https://doi.org/10.1038/s41467-024-44899-8>.

8. H. Chen, R.-T. Gao, F. Yang, L. Wu, and L. Wang, "High-Selectivity Electrocatalytic Oxidation of Glycerol to C₃-Glyceraldehyde Coupled to H₂O₂ Production," *Angewandte Chemie International Edition* 64 (2025): 202514737, <https://doi.org/10.1002/anie.202514737>.
9. S. Siahrostami, A. Verdaguer-Casadevall, M. Karamad, et al., "Enabling Direct H₂O₂ Production through Rational Electrocatalyst Design," *Nature Materials* 12 (2013): 1137–1143, <https://doi.org/10.1038/nmat3795>.
10. J. Wang, S. Huang, Y. Xu, et al., "Orbital-coupled Dual-atom Ni-Sn in Conductive Metal-organic Framework Enables Integrated Two-electron ORR and WOR for Efficient H₂O₂ Electrosynthesis," *Applied Catalysis B: Environment and Energy* 384 (2026): 126177, <https://doi.org/10.1016/j.apcatb.2025.126177>.
11. P. Cao, X. Quan, X. Nie, et al., "Metal Single-site Catalyst Design for Electrocatalytic Production of Hydrogen Peroxide at Industrial-relevant Currents," *Nature Communications* 14 (2023): 172, <https://doi.org/10.1038/s41467-023-35839-z>.
12. Z.-W. Chai, J.-R. Huang, M.-D. Zhang, X.-M. Chen, and P.-Q. Liao, "Efficient Capture and Conversion of O₂ into Salt-free Medical-grade H₂O₂ via a Metal-organic Framework with Enzyme-mimicking Dicopper Sites in a Pressurized Electrolyzer," *Science China Chemistry* 68 (2025): 6474–6482, <https://doi.org/10.1007/s11426-025-2850-x>.
13. M.-D. Zhang, J.-R. Huang, C.-P. Liang, X.-M. Chen, and P.-Q. Liao, "Continuous Electrosynthesis of Pure H₂O₂ Solution with Medical-Grade Concentration by a Conductive Ni-Phthalocyanine-Based Covalent Organic Framework," *Journal of the American Chemical Society* 146 (2024): 31034–31041, <https://doi.org/10.1021/jacs.4c10675>.
14. Z. You, B. Wang, Z. Zhao, et al., "Metal-Free Carbon-Based Covalent Organic Frameworks with Heteroatom-Free Units Boost Efficient Oxygen Reduction," *Advanced Materials* 35 (2023): 2209129, <https://doi.org/10.1002/adma.202209129>.
15. X. Yang, Q. Xu, W. Wei, and G. Zeng, "Linkages Chemistry of Covalent Organic Frameworks in Photocatalysis and Electrocatalysis," *Angewandte Chemie International Edition* 64 (2025): 202504355, <https://doi.org/10.1002/anie.202504355>.
16. J. Liu, W. Zhang, J. Shen, L. Feng, Y. Yao, and Q. Peng, "Rational Design Two- or Four-Electron Reaction Pathway Covalent Organic Frameworks for Efficient and Selective Electrocatalytic Hydrogen Peroxide Production," *Angewandte Chemie International Edition* 64 (2025): 202424720, <https://doi.org/10.1002/anie.202424720>.
17. R. Bao, Z. Xiang, Z. Qiao, et al., "Designing Thiophene-Enriched Fully Conjugated 3D Covalent Organic Framework as Metal-Free Oxygen Reduction Catalyst for Hydrogen Fuel Cells," *Angewandte Chemie International Edition* 62 (2023): 202216751, <https://doi.org/10.1002/anie.202216751>.
18. W. Peng, J. Liu, X. Liu, et al., "Facilitating Two-electron Oxygen Reduction with Pyrrolic Nitrogen Sites for Electrochemical Hydrogen Peroxide Production," *Nature Communications* 14 (2023): 4430, <https://doi.org/10.1038/s41467-023-40118-y>.
19. F. Xiang, X. Zhao, J. Yang, et al., "Enhanced Selectivity in the Electroproduction of H₂O₂ via F/S Dual-Doping in Metal-Free Nanofibers," *Advanced Materials* 35 (2023): 2208533, <https://doi.org/10.1002/adma.202208533>.
20. P. Côté, A. I. Benin, N. W. Ockwig, M. O'Keeffe, A. J. Matzger, and O. M. Yaghi, "Porous, Crystalline, Covalent Organic Frameworks," *Science* 310 (2005): 1166–1170.
21. R.-B. L. Yu, G. Xu, Z.-H. Fu, et al., "Linkage Conversions in Single-crystalline Covalent Organic Frameworks," *Nature Chemistry* 16 (2024): 114–121, <https://doi.org/10.1038/s41557-023-01334-7>.
22. W. Li, Z. Bie, C. Zhang, et al., "Combinatorial Synthesis of Covalent Organic Framework Particles with Hierarchical Pores and Their Catalytic Application," *Journal of the American Chemical Society* 145 (2023): 19283–19292, <https://doi.org/10.1021/jacs.3c04995>.
23. S. Fu, E. Jin, H. Hanayama, et al., "Outstanding Charge Mobility by Band Transport in Two-Dimensional Semiconducting Covalent Organic

- Frameworks,” *Journal of the American Chemical Society* 144 (2022): 7489–7496, <https://doi.org/10.1021/jacs.2c02408>.
24. J. Qiu, K. Meng, Y. Zhang, et al., “COF/in 2S 3 S-Scheme Photocatalyst with Enhanced Light Absorption and H₂O₂ -Production Activity and FST-A Investigation,” *Advanced Materials* 36 (2024): 2400288, <https://doi.org/10.1002/adma.202400288>.
25. X. Cui, M. Wu, X. Liu, et al., “Engineering Organic Polymers as Emerging Sustainable Materials for Powerful Electrocatalysts,” *Chemical Society Reviews* 53 (2024): 1447–1494, <https://doi.org/10.1039/D3CS00727H>.
26. H. Peng, S. Huang, V. Montes-García, et al., “Supramolecular Engineering of Cathode Materials for Aqueous Zinc-Ion Energy Storage Devices: Novel Benzothiadiazole Functionalized Two-Dimensional Olefin-Linked COFs,” *Angewandte Chemie International Edition* 62 (2023): 202216136, <https://doi.org/10.1002/anie.202216136>.
27. S. Huang, K. Chen, and T.-T. Li, “Porphyrin and Phthalocyanine Based Covalent Organic Frameworks for Electrocatalysis,” *Coordination Chemistry Reviews* 464 (2022): 214563, <https://doi.org/10.1016/j.ccr.2022.214563>.
28. Y. Su, K.-I. Otake, J.-J. Zheng, S. Horike, S. Kitagawa, and C. Gu, “Separating Water Isotopologues Using Diffusion-Regulatory Porous Materials,” *Nature* 611 (2022): 289–294, <https://doi.org/10.1038/s41586-022-05310-y>.
29. N. Keller and T. Bein, “Optoelectronic Processes in Covalent Organic Frameworks,” *Chemical Society Reviews* 50 (2021): 1813–1845.
30. Q. Chen, D.-H. Si, Q.-J. Wu, R. Cao, and Y.-B. Huang, “Engineering Copper-Based Covalent Organic Framework Microenvironments to Enable Efficient CO₂ Electrorreduction with Tunable Ethylene/Methane Switch,” *Advanced Functional Materials* 34 (2024): 2315368, <https://doi.org/10.1002/adfm.202315368>.
31. H. Guo, D.-H. Si, H.-J. Zhu, Z.-A. Chen, R. Cao, and Y.-B. Huang, “Boosting CO₂ Electrorreduction over a Covalent Organic Framework in the Presence of Oxygen,” *Angewandte Chemie International Edition* 63 (2024): 202319472, <https://doi.org/10.1002/anie.202319472>.
32. Y. Xu, J.-P. Dong, L. Wang, et al., “3D Cluster-Based Covalent Organic Framework for Efficient CO₂ Photoreduction,” *Angewandte Chemie International Edition* 64 (2025): 202501391, <https://doi.org/10.1002/anie.202501391>.
33. H.-Y. Chen, J.-R. Huang, J.-C. Liu, N.-Y. Huang, X.-M. Chen, and P.-Q. Liao, “Integration of Plasmonic Ag(I) Clusters and Fe(II) Porphyrinates into Metal–Organic Frameworks for Efficient Photocatalytic CO₂ Reduction Coupling with Photosynthesis of Pure H₂O₂,” *Angewandte Chemie International Edition* 63 (2024): 202412553, <https://doi.org/10.1002/anie.202412553>.
34. M. Liu, D. Jiang, Y. Fu, et al., “Modulating Skeletons of Covalent Organic Framework for High-Efficiency Gold Recovery,” *Angewandte Chemie International Edition* 63 (2024): 202317015, <https://doi.org/10.1002/anie.202317015>.
35. S. Huang, B. Zhang, D. Wu, et al., “Linkage Engineering in Covalent Organic Frameworks as Metal-free Oxygen Reduction Electrocatalysts for Hydrogen Peroxide Production,” *Applied Catalysis B: Environmental* 340 (2024): 123216, <https://doi.org/10.1016/j.apcatb.2023.123216>.
36. Z. Zhang, Z. Zhang, C. Chen, et al., “Single-atom Platinum with Asymmetric Coordination Environment on Fully Conjugated Covalent Organic Framework for Efficient Electrocatalysis,” *Nature Communications* 15 (2024): 2556, <https://doi.org/10.1038/s41467-024-46872-x>.
37. T. Zhang, Z. Jiang, and A. M. Rappe, “Hydrogenation of Covalent Organic Framework Induces Conjugated π Bonds and Electronic Topological Transition to Enhance Hydrogen Evolution Catalysis,” *Journal of the American Chemical Society* 146 (2024): 15488–15495, <https://doi.org/10.1021/jacs.4c03973>.
38. X. Yang, Y. Fu, M. Liu, et al., “Solvent Effects on Metal-Free Covalent Organic Frameworks in Oxygen Reduction Reaction,” *Angewandte Chemie International Edition* 63 (2024): 202319247, <https://doi.org/10.1002/anie.202319247>.
39. X. Yang, Q. An, X. Li, et al., “Charging Modulation of the Pyridine Nitrogen of Covalent Organic Frameworks for Promoting Oxygen Reduction Reaction,” *Nature Communications* 15 (2024): 1889, <https://doi.org/10.1038/s41467-024-46291-y>.
40. X. Li, S. Yang, M. Liu, et al., “Catalytic Linkage Engineering of Covalent Organic Frameworks for the Oxygen Reduction Reaction,” *Angewandte Chemie International Edition* 62 (2023): 202304356, <https://doi.org/10.1002/anie.202304356>.
41. H.-L. Zhu, J.-R. Huang, F. Zhang, P.-Q. Liao, and X.-M. Chen, “Observation of O₂ Molecules Inserting into Fe–H Bonds in a Ferrous Metal–Organic Framework,” *Journal of the American Chemical Society* 147 (2025): 4595–4601, <https://doi.org/10.1021/jacs.4c16736>.
42. R. Wang, H. Zhou, Q. Fang, Z. Su, and S. Qiu, “Recent Advances in Covalent Organic Framework-based Catalysts for the Electrosynthesis of Hydrogen Peroxide,” *Chemical Communications* 61 (2025): 15346–15361, <https://doi.org/10.1039/D5CC03377B>.
43. R. Qiao, J. Wang, H. Hu, and S. Lu, “Covalent Organic Frameworks Based Electrocatalysts for Two-Electron Oxygen Reduction Reaction: Design Principles, Recent Advances, and Perspective,” *Molecules (Basel, Switzerland)* 29 (2024): 2563, <https://doi.org/10.3390/molecules29112563>.
44. Y. Zhang, Z. Qiao, R. Zhang, et al., “Multicomponent Synthesis of Imidazole-Linked Fully Conjugated 3D Covalent Organic Framework for Efficient Electrochemical Hydrogen Peroxide Production,” *Angewandte Chemie International Edition* 62 (2023): 202314539, <https://doi.org/10.1002/anie.202314539>.
45. G. Liu, M. Lin, Y. Xing, et al., “Oligoether Chain Engineering in Covalent Organic Frameworks: Enhancing Transport Pathways and Oxygen Reduction Activity for Efficient Electrocatalytic Hydrogen Peroxide Production,” *Angewandte Chemie International Edition* 64 (2025): 202500945, <https://doi.org/10.1002/anie.202500945>.
46. S. An, X. Li, S. Shang, et al., “One-Dimensional Covalent Organic Frameworks for the 2e[−] – Oxygen Reduction Reaction,” *Angewandte Chemie International Edition* 62 (2023): 202218742, <https://doi.org/10.1002/anie.202218742>.
47. M. Martínez-Fernández, E. Martínez-Periñán, A. de la Peña Ruigómez, et al., “Scalable Synthesis and Electrocatalytic Performance of Highly Fluorinated Covalent Organic Frameworks for Oxygen Reduction,” *Angewandte Chemie International Edition* 62 (2023): 202313940, <https://doi.org/10.1002/anie.202313940>.
48. R. Wang, Z. Zhang, H. Zhou, et al., “Structural Modulation of Covalent Organic Frameworks for Efficient Hydrogen Peroxide Electrocatalysis,” *Angewandte Chemie International Edition* 63 (2024): 202410417, <https://doi.org/10.1002/anie.202410417>.
49. Y. Chen, C. Zhen, Y. Chen, et al., “Oxygen Functional Groups Regulate Cobalt-Porphyrin Molecular Electrocatalyst for Acidic H₂O₂ Electrosynthesis at Industrial-Level Current,” *Angewandte Chemie International Edition* 63 (2024): 202407163, <https://doi.org/10.1002/anie.202407163>.
50. J. Gao, H. B. Yang, X. Huang, et al., “Enabling Direct H₂O₂ Production in Acidic Media through Rational Design of Transition Metal Single Atom Catalyst,” *Chemistry (Weinheim An Der Bergstrasse, Germany)* 6 (2020): 658–674, <https://doi.org/10.1016/j.chempr.2019.12.008>.
51. Y. Wang, Y. Zhou, Y. Feng, and X.-Y. Yu, “Synergistic Electronic and Pore Structure Modulation in Open Carbon Nanocages Enabling Efficient Electrocatalytic Production of H₂O₂ in Acidic Medium,” *Advanced Functional Materials* 32 (2022): 2110734, <https://doi.org/10.1002/adfm.202110734>.
52. C. Zhang, W. Shen, K. Guo, M. Xiong, J. Zhang, and X. Lu, “A Pentagonal Defect-Rich Metal-Free Carbon Electrocatalyst for Boosting Acidic O₂ Reduction to H₂O₂ Production,” *Journal of the American Chemical Society* 145 (2023): 11589–11598, <https://doi.org/10.1021/jacs.3c00689>.
53. L. Wei, T. Sun, Z. Shi, et al., “Guest-adaptive Molecular Sensing in a Dynamic 3D Covalent Organic Framework,” *Nature Communications* 13 (2022): 7936, <https://doi.org/10.1038/s41467-022-35674-8>.

54. Y.-X. Ma, Z.-J. Li, L. Wei, S.-Y. Ding, Y.-B. Zhang, and W. Wang, "A Dynamic Three-Dimensional Covalent Organic Framework," *Journal of the American Chemical Society* 139 (2017): 4995–4998, <https://doi.org/10.1021/jacs.7b01097>.
55. X. Liu, J. Li, B. Gui, et al., "A Crystalline Three-Dimensional Covalent Organic Framework with Flexible Building Blocks," *Journal of the American Chemical Society* 143 (2021): 2123–2129, <https://doi.org/10.1021/jacs.0c12505>.
56. Y. Chen, Z.-L. Shi, L. Wei, et al., "Guest-Dependent Dynamics in a 3D Covalent Organic Framework," *Journal of the American Chemical Society* 141 (2019): 3298–3303, <https://doi.org/10.1021/jacs.8b13691>.

Supporting Information

Additional supporting information can be found online in the Supporting Information section.

Supporting File: adfm75407-sup-0001-SuppMat.docx.